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Lab class Task

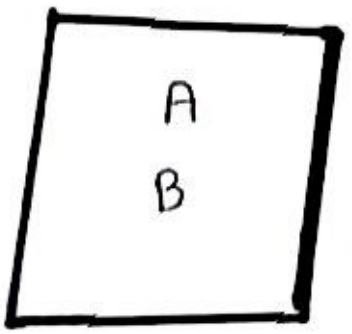
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Task 1:-

Run these program and observe their output:

```
#include<iostream>
using namespace std;
 int main ( )
 {
Char = Letter ;
Letter = 'A' ;
Cout<< Letter<< 'n' ;
Cin>>
 Letter = 'B' ;
 Cout<< Letter<< 'n' ;
   Cin>> Letter ;
   return 0 ;
 }
```

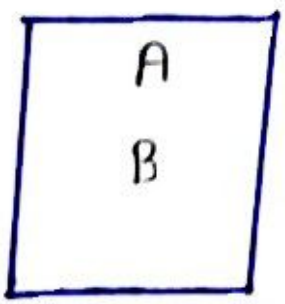
output:-



Run these program and observe their output:-

```
#include<iostream>
using namespace std;
 int main()
{
char letter;
letter = 65;
cout<<letter<<'\n';
 letter = 66;
cout<<letter<<'\n';
   cin>> letter;
   return 0;
}
```

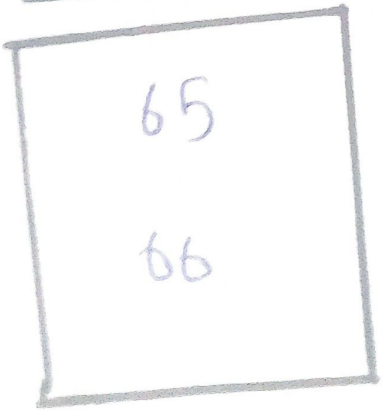
output:-



Run these program and observe their output.

```
#include <iostream>
using namespace std;
int main()
{
    int letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    cin >> letter;
    return 0;
}
```

output



Task 2:

Take two integers as input from the user apply athematic operations on them (+, -, *, /) as print them on screen.

```
# include <iostream>
using namespace std;
int main()
{
  int a, b;
cout<< "Enter two numbers:";
 cin>> a >>b;
cout<< " Sum=" << a+b << endl;
cout<< " Sub=" << a-b <<endl;
cout<< "Mul=" <<a*b<<endl;
cout<< "Div=" <<a/b<<endl;
 return 0;
}
```

Task3:

write the output of the following code.

```
#include "stdafx.h"
#include<iostream>
using namespace std;
int main()
{
 int val1;
 int val2;
 int val3;
cout<<"please enter your 3 numbers:";
cin >> val1>> val2>> val3;
cout<< val1<<"\n"<< val2<<"\n"<<val3<<"\n";
cout<<val3<<"\n"<< val2<<"\n"<<val1<<"\n";
 system("pause");
 return 0;
}
```

output

```
1
2
3
3
2
1
```

Task - 4

Write a C++ Program to calculate the distance between the two points.

Note: x_1, y_1, x_2, y_2 are all double values.

```
#include<iostream>
#include<cmath>
using namespace std;
  int main()
  {
      double x1, y1, x2, y2;
       double distance;
    Cout<<"Enter x1:";
    Cin>>x1;
    Cout<<"Enter y1:";
    Cin>>y1;
    Cout<<"Enter x2:";
    Cin>>x2;
    Cout<<"Enter y2:";
    Cin>>y2;
   distance = sqrt(pow(x2-x1,2)+ pow(y2-y1,2));
    Cout<<"Distance between two points is:"<<distance<<endl;
     return 0;
  }
```

Output

```
Enter x1: 1
Enter y1: 2
Enter x2: 4
Enter y2: 6
Distance between two points is 5
```

Task - 5

(1) Calculate the area of circle (area = pie x r^2)

```
#include<iostream>
using namespace std;
int main()
{
    double r, area;
    const double pie= 3.14159;
    cout<<"Enter radius of circle:";
     cin >> r ;
    area = pie * r * r ;
    cout<<"Area of a circle:"<<area<<endl;
      return 0;
}
```

Output

```
Enter radius of a circle:5
Area of a circle:78.5397
```

Task - 6

Total Purchase A customer in a store is purchasing five items. The prices of the five items are:

Price of item 1 = $ 12.95
Price of item 2 = $ 24.95
Price of item 3 = $ 6.95
Price of item 4 = $ 14.95
Price of item5 = $ 3.95

Write a Program that holds the prices of the fives items in five variables. Display each items Price, the subtotal of the sale, the amount of sales tax and the total. Assume the sales tax is 6%.

```
#include <iostream>
using namespace std;
int main ()
{
    float item 1 = 12.95f;
    float item 2 = 24.95f;
    float item 3 = 6.95;
    float item4 = 14.95;
    float item5 = 3.95f
```

```
float Subtotal, SalesTax , Total;
const float Tax_Rate = 0.06f;
Subtotal = item1 + item2 + item3 + item4 + item5;
SalesTax = Subtotal * Tax_Rate;
Total = Subtotal + SalesTax;
Cout<<"Price of item1: $"<<item1<<endl;
cout<<"Price of item2: $"<<item2<<endl;
cout<<"Price of item3: $"<<item3<<endl;
cout<<"Price of item4: $"<<item4<<endl;
cout<<"Price of item5: $"<<item5<<endl;

cout<<"Subtotal of the sale: $"<<Subtotal<<endl;
cout<<"Amount of SaleTax (6%): $"<<Sale tax<<endl;
cout<<"Total: $"<<Total<<endl;
return 0;
}
```

output

```
Price of item1: $12.95
Price of item2: $24.95
Price of item3: $6.95
Price of item4: $1495
Price of item5: $ 3.95
Subtotal of the sale: 63.75
Amount of salesTax: 3.825
Total: $ 67.575
```

```cpp
#include<iostream>
using namespace std;
int main(){
    char letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    return 0;
}
```

```
A
B

[Program finished]
```



```cpp
#include<iostream>
using namespace std;
int main()
{
    char letter;
    letter = 65;
    cout << letter << '\n';
    letter = 66;
    cout << letter << '\n';
    return 0;
}
```

```
A
B

Program finished
```

```cpp
#include<iostream>
using namespace std;
int main(){
    int letter;
    letter = 'A';
    cout << letter << '\n';
    letter = 'B';
    cout << letter << '\n';
    return 0;
}
```

```
65
66

[Program finished]
```

```cpp
#include<iostream>
using namespace std;
int main(){
    int a, b;
    cout << "Enter two integers: ";
    cin >> a >> b;

    cout << "Addition: " << a+b << endl;
    cout << "Subtraction: " << a-b << endl;
    cout << "Multiplication: " << a*b << endl;
    cout << "Division: " << a/b << endl;
    return 0;
}
```

```
Enter two integers: 5 10
Addition: 15
Subtraction: -5
Multiplication: 50
Division: 0

[Program finished]
```



```cpp
#include<iostream>
using namespace std;
int main(){
    int val1, val2, val3;
    cout << "Please enter your 3 numbers: ";
    cin >> val1 >> val2 >> val3;

    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";
    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";
    return 0;
}
```

```
Please enter your 3 numbers: 1 2 3
1
2
3
3
2
1

[Program finished]
```

```cpp
#include<iostream>
#include<cmath>
using namespace std;
int main(){
    double x1, y1, x2, y2, d;
    cout << "Enter x1 y1: ";
    cin >> x1 >> y1;
    cout << "Enter x2 y2: ";
    cin >> x2 >> y2;

    d = sqrt( pow(x2-x1, 2) + pow(y2-y1, 2) );

    cout << "Distance = " << d << endl;
    return 0;
}
```

```
Enter x1 y1: 2 3
Enter x2 y2: 5 7
Distance = 5

[Program finished]
```

```cpp
#include<iostream>
using namespace std;
int main(){
    double r, area;
    const double PI = 3.14159;
    cout << "Enter radius: ";
    cin >> r;
    area = PI * r * r;
    cout << "Area of Circle = " << area << endl;
    return 0;
}
```

```
Enter radius: 5
Area of Circle = 78.5397

[Program finished]
```

```cpp
#include<iostream>
using namespace std;
int main(){
    double length, width, area;
    cout << "Enter length and width: ";
    cin >> length >> width;
    area = length * width;
    cout << "Area of Rectangle = " << area << endl;
    return 0;
}
```

```
Enter length and width: 10 4
Area of Rectangle = 40

Program finished
```

```cpp
#include<iostream>
using namespace std;
int main(){
    double base, height, area;
    cout << "Enter base and height: ";
    cin >> base >> height;
    area = base * height * 0.5;
    cout << "Area of Triangle = " << area << endl;
    return 0;
}
```

```
Enter base and height: 6 8
Area of Triangle = 24

[Program finished]
```

```
1  #include<iostream>
2  using namespace std;
3  int main(){
4      double item1=12.95, item2=24.95, item3=6.95,
   item4=14.95, item5=3.95;
5      double subtotal, tax, total;
6
7      subtotal = item1+item2+item3+item4+item5;
8      tax = subtotal * 0.06;
9      total = subtotal + tax;
10
11     cout << "Price of item 1 = $" << item1 << endl;
12     cout << "Price of item 2 = $" << item2 << endl;
13     cout << "Price of item 3 = $" << item3 << endl;
14     cout << "Price of item 4 = $" << item4 << endl;
15     cout << "Price of item 5 = $" << item5 << endl;
16     cout << "Subtotal = $" << subtotal << endl;
17     cout << "Sales Tax (6%) = $" << tax << endl;
18     cout << "Total = $" << total << endl;
19     return 0;
20 }
```

```
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95
Subtotal = $63.75
Sales Tax (6%) = $3.825
Total = $67.575

[Program finished]
```

```cpp
#include<iostream>
using namespace std;
int main(){
    int num;
    cout << "Enter a five-digit integer: ";
    cin >> num;

    int d1 = num/10000;
    int d2 = (num/1000)%10;
    int d3 = (num/100)%10;
    int d4 = (num/10)%10;
    int d5 = num%10;

    cout << d1 << "   " << d2 << "   " << d3 << "   " << d4
<< "   " << d5 << endl;
    return 0;
}
```

```
Enter a five-digit integer: 12345
1  2  3  4  5

[Program finished]
```

```cpp
#include<iostream>
using namespace std;
int main(){
    int var1, var2, temp;
    cout << "Enter var1 and var2: ";
    cin >> var1 >> var2;

    cout << "Before Swap: var1=" << var1 << " var2=" << var2 << endl;

    temp = var1;
    var1 = var2;
    var2 = temp;

    cout << "After Swap: var1=" << var1 << " var2=" << var2 << endl;
    return 0;
}
```

```
Enter var1 and var2: 10 20
Before Swap: var1=10 var2=20
After Swap: var1=20 var2=10

Program finished
```

```cpp
#include<iostream>
using namespace std;
int main(){
    char small;
    cout << "Enter small alphabet (a-z): ";
    cin >> small;

    char capital = small - 32; // ASCII trick, a=97, A=65 difference 32

    cout << "Upper case = " << capital << endl;
    return 0;
}
```

```
Enter small alphabet (a-z): b
Upper case = B

[Program finished]
```

